

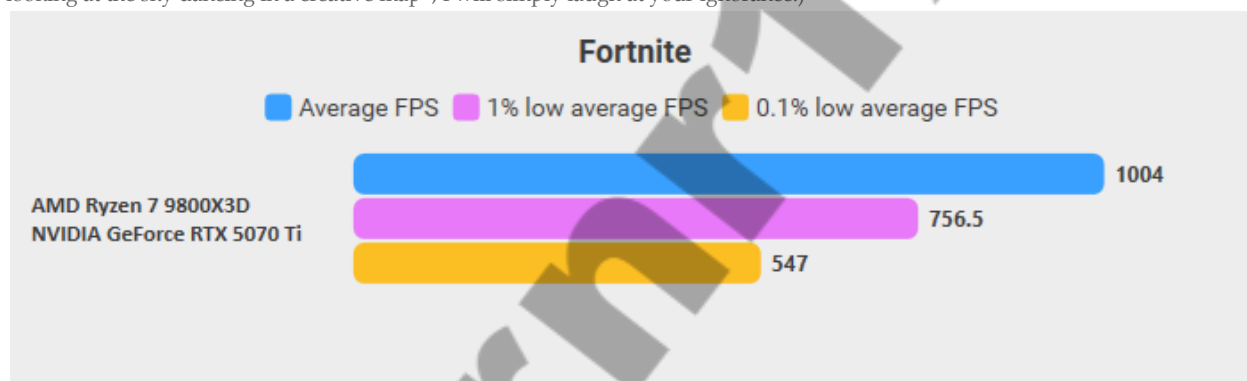
# Turnr1x AM5 X3D Bios Guide

## Overview

This is a guide for AM5 X3D bios settings that I have concluded provide the best performance. For my specific case I am on an X3D (9800X3D) chip and these settings yielded the best results for my system. While this guide works perfectly fine in 99.99% of cases, **DO NOT** mass apply the settings, I do not recommend it. Each setting should be tested independently on your system to make sure they provide better performance and do not affect your system negatively in any way. I **am not liable** for any damage done to your system.

## Benchmark Results

(These are placed at the top just to show these settings do in fact have a positive impact. Do not DM me saying "I get 1200fps looking at the sky dancing in a creative map", I will simply laugh at your ignorance.)



(My results from the Fortnite minecart benchmark at 1080p. Results are the aggregate of 5 total runs. I do not have my old picture from CS2 bench, but P1 lows were slightly over 400 with bots at 1280x960.)

## Settings

ACP Power Gating = Disabled  
ACP Clock Gating = Disabled  
Unused GPP Clocks Off = Disabled  
PME Turn Off Support = Disabled  
S3 PCIe Save Restore Mode = Disabled or Mode 2 (TEST)  
Global C-state Control = Enabled  
DF Cstates = Disabled  
Disable DF to external downstream IP Sync Flood Propagation = Disabled  
Disable DF Sync flood propagation = Disabled  
Freeze DF module queues on error = Disabled  
PSPP Policy = Disabled  
Core Watchdog Timer = Disabled  
Core Watchdog Timer Severity = No error  
Power Supply Control = Typical Current Idle  
Prochot VRM Throttling = Disabled  
ACPI\_CST C1 Declaration = Disabled  
ACPI Standby State = Disabled  
ASPM Control for CPU PCIe = Disabled

PAUSE Delay = Disabled or Auto (TEST)  
Streaming Stores Control = Enabled  
OPCache Control = Enabled  
FCH Spread Spectrum = Disabled  
Core Tuning Configuration for Gaming = Level 2 or Legacy (Level 2 yielded the best results for me)  
SVM Enable = Disabled  
IOMMU = Disabled (UNLESS NEEDED FOR CERTAIN GAME REQUIREMENTS)  
REP-MOV/STOS Streaming = Enabled  
Enhanced REP MOVSB/STOSB = Enabled  
Fast Short REP MOVSB = Enabled  
L3 DFLS Stretch Mode = Disabled  
L1/L2 Stream HW Prefetcher = Enabled  
L1 Stride Prefetcher = Enabled  
L1 Region Prefetcher = Enabled  
L1 Burst Prefetch Mode = Enabled  
L2 Up-Down Prefetcher = Enabled  
CPU Speculative Store Modes = More Speculative  
PSS Support = Disabled  
PS2 Devices Support = Disabled  
ACPI SRAT L3 Cache As Numa Domain = Disabled  
Latency Under Load = Disabled or Auto (TEST)  
iGPU/Integrated Graphics = Disabled (Unless you use it)  
Any Motherboard Driver Installer/ App = Disabled  
Discrete GPU's USB Port = Disabled  
USB power delivery in Soft Off state (S5) = Disabled  
TSME = Disabled  
SMEE = Disabled  
Data scramble = Disabled  
Memory P-state = Disabled  
Memory Fast Boot = Disabled  
Memory Context Restore = Disabled  
Power Down = Disabled  
ECC = Disabled  
PCIe All Port ECRC = Disabled  
PCIe loopback Mode = Disabled  
Persistence mode for legacy endpoints = Disabled  
NPU Deep Sleep Enable = Disabled  
Advanced Error Reporting = Disabled  
Platform First Error Handling = Disabled  
MCA Error Thresh = False  
MCA FruText = False  
SMU and PSP Debug Mode = Disabled  
Action on BIST Failure = Dp nothing  
Log Transparent Errors = Disabled  
BankSwapMode = Swap APU  
MBIST Enable = Disabled  
MBIST Aggressors = Disabled  
MBIST Per Bit Slave Die Reporting = Disabled  
Aggressor Channel = Disabled  
Onboard Wifi = Disabled (Unless you use it)  
Onboard Bluetooth = Disabled (Unless you use it)

—Disable ALL of the below D3 settings—

ACPI D3 Support  
Internal PCIe GPP 0 D3  
SOC GPU D3  
SOC HD AUDIO D3  
SOC AZALIA D3  
SOC USB 3.1 D3  
SOC ACP D3  
SOC USB 2.0 D3  
SOC USB 3.1 for USB 4 D3  
SOC USB D3  
Internal USB4 PCIe Tunneling D3

### ***Conclusion***

THANK YOU for checking out my guide, I greatly appreciate it! I hope it has provided your system with better performance and made gaming more enjoyable. There are obviously many more settings that can be changed to increase performance but a lot are system dependent and require too much testing to just slap on a guide and call it “universal”. If you like this drop a follow on my socials @Turnr1x on pretty much everything you can think of. If this gets the right support I hope to do many more things like this in the future. If you have any questions or want some more advice, reach out to me on discord or twitter. Thanks again!